

PERFUME SHOP MANAGEMENT SYSTEM

1. Customer

Primary Key (PK)

- Customer_ID

Attributes

- Customer_ID (PK)
- Customer_Name
- Gender
- Phone_Number
- Email
- Address
- Date_of_Birth
- Username
- Password

Relationship

- Customer places Orders.
 - Customer makes Payments.
 - Customer writes Reviews.
-

2. Perfume

Primary Key (PK)

- Perfume_ID

Attributes

- Perfume_ID (PK)
- Perfume_Name

- Brand_ID (FK)
- Category_ID (FK)
- Price
- Quantity_In_Stock
- Volume (30ml, 50ml, 100ml)
- Fragrance_Type
- Description
- Manufacture_Date
- Expiry_Date

Relationship

- Perfume belongs to one Brand.
 - Perfume belongs to one Category.
 - Perfume appears in many Orders.
 - Perfume is supplied by Suppliers.
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3. Brand

Primary Key

- Brand_ID

Attributes

- Brand_ID (PK)
- Brand_Name
- Country
- Brand_Description

Relationship

- One Brand manufactures many Perfumes.
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4. Category

Primary Key

- Category_ID

Attributes

- Category_ID (PK)
- Category_Name
- Description

Example Categories

- Men
- Women
- Unisex
- Luxury
- Floral
- Woody

Relationship

- One Category contains many Perfumes.
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5. Supplier

Primary Key

- Supplier_ID

Attributes

- Supplier_ID (PK)
- Supplier_Name
- Phone_Number
- Email
- Address

Relationship

- Supplier supplies many Perfumes.
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6. Employee

Primary Key

- Employee_ID

Attributes

- Employee_ID (PK)
- Employee_Name
- Position
- Phone_Number
- Email
- Salary

Relationship

- Employee manages Orders.
 - Employee manages Inventory.
-

7. Order

Primary Key

- Order_ID

Foreign Key

- Customer_ID

Attributes

- Order_ID (PK)
- Customer_ID (FK)
- Order_Date

- Order_Status
- Total_Amount
- Delivery_Address

Relationship

- Customer places many Orders.
 - One Order contains many Perfumes.
 - One Order has one Payment.
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8. Order Details

Primary Key

- Order_Detail_ID

Foreign Keys

- Order_ID
- Perfume_ID

Attributes

- Order_Detail_ID (PK)
- Order_ID (FK)
- Perfume_ID (FK)
- Quantity
- Unit_Price
- Total_Price

Relationship

- Resolves the Many-to-Many relationship between Orders and Perfumes.
-

9. Payment

Primary Key

- Payment_ID

Foreign Key

- Order_ID

Attributes

- Payment_ID (PK)
- Order_ID (FK)
- Payment_Method
- Payment_Date
- Payment_Status
- Amount

Relationship

- One Order has one Payment.
-

10. Inventory

Primary Key

- Inventory_ID

Foreign Key

- Perfume_ID

Attributes

- Inventory_ID (PK)
- Perfume_ID (FK)
- Stock_Available
- Reorder_Level
- Last_Updated

Relationship

- Inventory stores stock details of Perfumes.

11. Review

Primary Key

- Review_ID

Foreign Keys

- Customer_ID
- Perfume_ID

Attributes

- Review_ID (PK)
- Customer_ID (FK)
- Perfume_ID (FK)
- Rating
- Review_Text
- Review_Date

Relationship

- Customer writes many Reviews.
- Perfume receives many Reviews.

Entity Relationships

Entity 1	Relationship	Entity 2	Cardinality
Customer	Places	Order	1 : M
Customer	Makes	Payment	1 : M
Customer	Writes	Review	1 : M
Brand	Manufactures	Perfume	1 : M
Category	Contains	Perfume	1 : M

Entity 1	Relationship	Entity 2	Cardinality
Supplier	Supplies	Perfume	M : N (<i>using Supplier_Perfume if needed</i>)
Order	Contains	Perfume	M : N (<i>resolved using Order_Details</i>)
Order	Has	Payment	1 : 1
Perfume	Stored In	Inventory	1 : 1
Employee	Manages	Inventory	1 : M
Employee	Processes	Order	1 : M
Perfume	Receives	Review	1 : M

Primary Keys

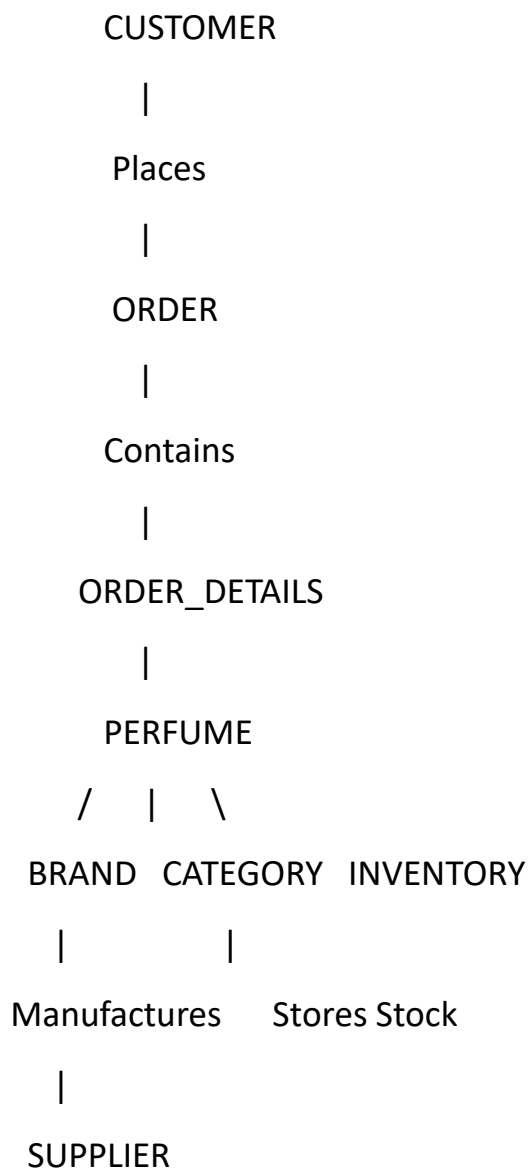
- Customer_ID
- Perfume_ID
- Brand_ID
- Category_ID
- Supplier_ID
- Employee_ID
- Order_ID
- Order_Detail_ID
- Payment_ID
- Inventory_ID
- Review_ID

Foreign Keys

- Customer_ID → Order
- Customer_ID → Review

- Brand_ID → Perfume
- Category_ID → Perfume
- Order_ID → Order_Details
- Perfume_ID → Order_Details
- Order_ID → Payment
- Perfume_ID → Inventory
- Perfume_ID → Review

ER Diagram Relationships



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PAYMENT

CUSTOMER ----- REVIEW ----- PERFUME

EMPLOYEE ----- Manages ----- INVENTORY

EMPLOYEE ----- Processes ----- ORDER

This design is **normalized (up to 3NF)** and is suitable for a **Perfume Shop Management System** mini-project. It provides a strong foundation for preparing an **ER diagram, relational schema, and SQL database implementation.**